

Please write clearly in block capitals.

Centre number

Candidate number

Surname

Forename(s)

A-level

ENVIRONMENTAL SCIENCE**Paper 1 (Generated practice paper - Set 31)**

Assessed: The physical environment, Energy resources, Pollution, Research methods

Time allowed: 3 hours**Materials**

For this paper you may use:

- a calculator.

Instructions

- Use black ink or black ball-point pen. Pencil should only be used for drawing.
- Fill in the boxes at the top of this page.
- Answer **all** questions 1 to 10 and **one** essay from question 11.
- You must answer the questions in the spaces provided. Do not write outside the box around each page or on blank pages.
- Do all rough work in this book. Cross through any work you do not want to be marked.

Information

- The marks for questions are shown in brackets.
- The maximum mark for this paper is 120.
- All questions should be answered in continuous prose.
- You will be assessed on your ability to:
 - use good English
 - organise information clearly
 - use specialist vocabulary where appropriate.

For Examiner's Use

Question	Mark
1	
2	
3	
4	
5	
6	
7	
8	
9	
10	
11	
TOTAL	

Answer all questions in the spaces provided.

0 1 . 1

Explain why oil and gas trapped in shale cannot be extracted by conventional drilling and describe how hydraulic fracturing (fracking) allows extraction.

[3 marks]

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0 1 . 2

Describe **four** environmental concerns about fracking and suggest how they may be reduced.

[5 marks]

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0 1 . 3

Explain why tar sands and oil shales are not included in most estimates of oil reserves despite containing more oil than conventional crude oil.

[2 marks]

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10

0 2

A student measured the mass of carbon stored per hectare in three land uses in the same region.

Table 1

Land use	Carbon in vegetation / t ha ⁻¹	Carbon in soil dead organic matter / t ha ⁻¹
Mature woodland	120	95
Permanent pasture	3	80
Arable (ploughed annually)	2	45

0 2

. 1

Calculate the total carbon lost per hectare when mature woodland is converted to arable farmland.

[1 mark]

Answer

t/ha

0 2

. 2

Explain why the soil carbon in arable land is so much lower than in pasture, even though the vegetation carbon is similar.

[2 marks]

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0 2

. 3

Suggest **two** ways the arable farmer could increase soil carbon storage.

[2 marks]

1

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2

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5

0 3 Figure 1 shows the daily deaths in London and the mean concentration of smoke in the air during the smog of December 1952.

Figure 1

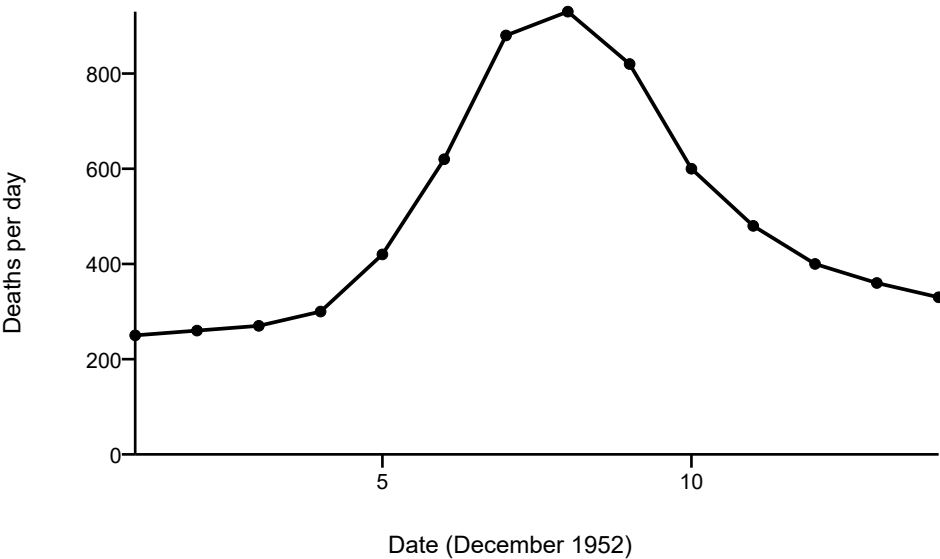


Table 1

Date	3 Dec	5 Dec	7 Dec	8 Dec	10 Dec	12 Dec
Smoke / mg m ⁻³	0.4	1.5	4.4	4.5	1.2	0.5

0 3 . 1 Describe the relationship between smoke concentration and daily deaths shown in Figure 1 and Table 1.

[2 marks]

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0 3 . 2 Calculate the percentage increase in daily deaths between 1 December and 8 December.

[2 marks]

Answer

%

03 . 3

Explain how the weather conditions in December 1952 caused the smog to form.

[3 marks]

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03 . 4

Explain why most of the deaths were among the elderly, the very young and people with existing respiratory disease.

[2 marks]

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03 . 5

State the legislation introduced in the UK in response to this smog.

[1 mark]

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10

0 4

Table 1 shows the effect of installing water meters on the mean daily water use of households in a UK town.

Table 1

	Unmetered households (fixed charge)	Metered households
Mean water use / dm ³ per person per day	150	125
Percentage with low-flush toilets	35	58
Percentage using a garden hosepipe weekly in summer	44	20

0 4

1

Calculate the percentage reduction in water use per person in metered households.

[1 mark]

Answer

%

0 4

2

Explain why metering reduces water use.

[2 marks]

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0 4

3

Describe **three** ways in which domestic water use can be reduced other than metering.

[3 marks]

1

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2

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3

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0 4 . 4

Explain why about 20% of the water intended for public supply in the UK never reaches consumers, and how this could be reduced.

[2 marks]

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0 4 . 5

Explain how drip irrigation reduces the water needed to grow a crop compared with spray irrigation.

[2 marks]

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10

0 5

Table 1 shows the bypass ratio and first year of use of jet engines used on four airliners.

Table 1

Engine	Aircraft	Bypass ratio	First use
RR Olympus	Concorde	0 : 1	Early 1950s
RR Conway	Boeing 707	0.25 : 1	Late 1950s
P&W JT9D	Boeing 747	5 : 1	1969
RR Trent 1000	Boeing 787	10 : 1	2011

0 5 . 1

Explain how a high bypass ratio reduces engine noise.

[3 marks]

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05 . 2

Describe **two** other features of aircraft design that reduce noise.

[2 marks]

1

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2

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05 . 3

Explain how each of the following airport operating procedures reduces noise nuisance.

[4 marks]

Constant descent angle

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Night flight restrictions

.....

Noise-based landing charges / quota count

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Multiple runways

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05 . 4

Explain why aircraft are noisier on take-off than on landing.

[1 mark]

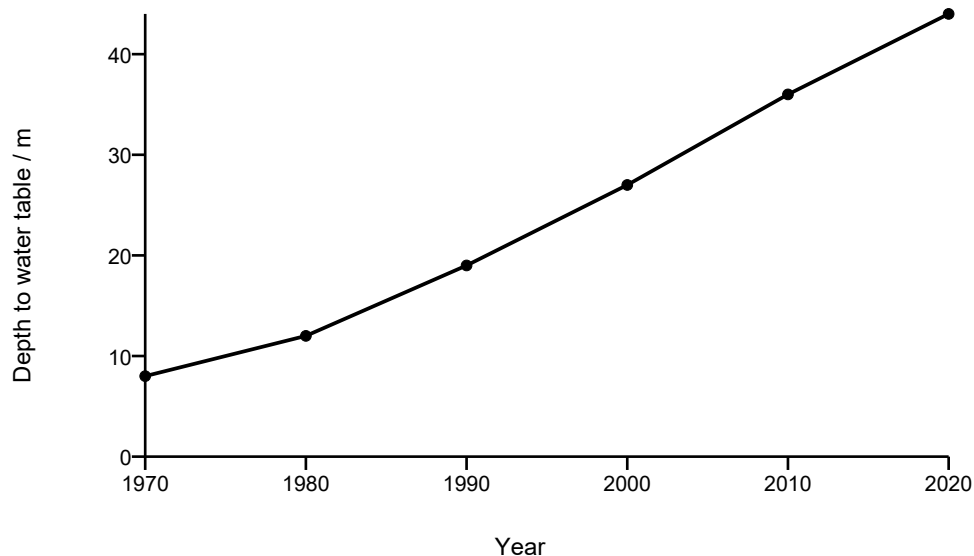
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10

0 6

Figure 1 shows the depth of the water table below the ground surface in an aquifer beneath a coastal farming region between 1970 and 2020.

Figure 1**0 6****. 1**

Calculate the mean rate at which the water table fell between 1970 and 2020.

[1 mark]**Answer**

m/yr

0 6**. 2**

Explain why the water table is falling.

[2 marks]

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0 6**. 3**

Explain the features a rock must have to form an aquifer.

[3 marks]

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06 . 4

Describe **four** consequences of over-exploiting this coastal aquifer.**[4 marks]**

1

2

3

4

10

07

Mercury released into a lake from a factory was in the inorganic form. Fish in the lake were later found to contain high concentrations of methyl mercury, an organic form.

07 . 1

Suggest how the inorganic mercury was converted to methyl mercury.

[2 marks]

07 . 2

Explain why methyl mercury is a more hazardous pollutant than inorganic mercury.

[3 marks]

07 . 3

Explain why the health risk from eating fish from the lake may remain for many years after the discharges stop.

[2 marks]

07 . 4

Suggest **three** factors that should be included in a Critical Pathway Analysis of a proposed mercury discharge.

[3 marks]

1

2

3

10

08 . 1

Explain why mineral extraction causes more land-use conflict than most other human activities.

[2 marks]

08 . 2

Describe how each of the following environmental impacts of an operating mine can be reduced.

[5 marks]

Dust

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Noise

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Turbid drainage water

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Acidic leachate from spoil

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Loss of amenity

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08 . 3

Explain how spoil heaps can be made stable and describe what happened at Aberfan in 1966.

[3 marks]

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10

09 . 1

Explain how ionising radiation damages living cells.

[3 marks]

09 . 2

Distinguish between acute and chronic effects of radiation exposure.

[2 marks]

09 . 3

Distinguish between **exposure** and **contamination**.

[2 marks]

09 . 4

A worker receives an absorbed dose of 0.002 Gy from alpha radiation.
Calculate the effective dose in sieverts.

[2 marks]

Answer

Sv

09 . 5

State what is measured by the becquerel.

[1 mark]

10

10 . 1

Neonicotinoids are the most widely used group of insecticides. Explain how the properties of neonicotinoids affect their impact on bee populations and describe the methods used to reduce this impact.

[9 marks]

Write an essay on one of the following topics.

$$\boxed{1} \boxed{1} . \boxed{1}$$

To what extent can renewable energy resources replace fossil fuels in meeting the energy needs of a modern society?

[25 marks]

OR

1 1 . 2

Discuss the extent to which the study of ozone depletion illustrates how scientific investigation can lead to the successful solution of an environmental problem.

[25 marks]

Indicate which question you have answered by writing its number, **11 . 1** or **11 . 2**, at the top of your answer.

This image shows a single sheet of white paper with horizontal ruling lines. The lines are evenly spaced and run across the width of the page. There is no text or other markings on the paper.

This image shows a full page of primary-ruled paper. It features a series of horizontal dashed lines spaced evenly down the page, designed for handwriting practice. A single solid line runs horizontally across the top of the page, serving as a baseline or header area. The rest of the page is white, providing ample space for writing.

END OF QUESTIONS